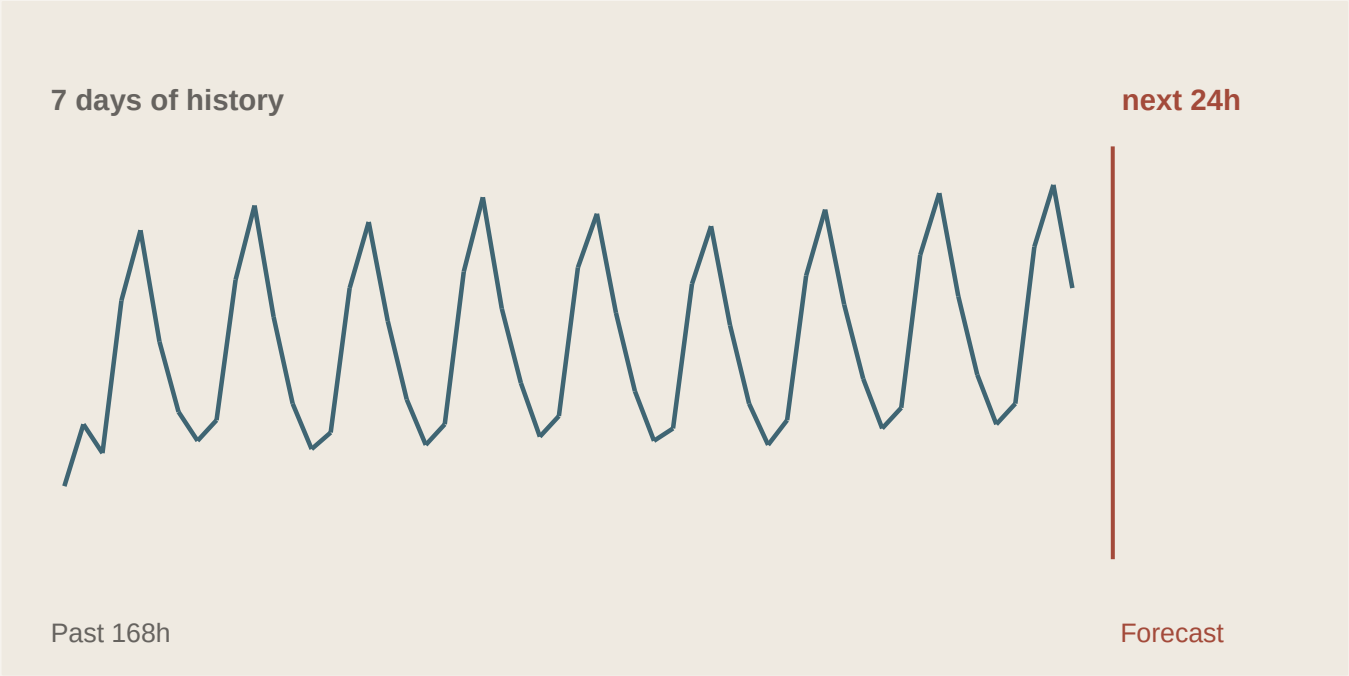
ELECTRICITY DEMAND FORECASTING

When is a more complex forecasting model actually worth it?

A practical comparison of a seasonal baseline, XGBoost, DLinear, LSTM, and PatchTST on the same 24-hour forecasting task.



The focus was not just the winning score, but whether the added complexity was worth it.

THE SETUP

Forecasting objective: use the previous 168 hours of demand to predict the next 24 hours.

About 50,400 hourly observations of German electricity demand. The model sees the previous 168 hours and predicts the next 24.

168 hours of history → **24-hour forecast**

Chronological split

2015–2018	2019	2020
Train	Validation	Test

No random split · **Train-only scaling** · **MAE · RMSE · WAPE**

The goal was to keep the comparison fair and avoid letting future information leak into training.

THE MODELS

Five ways to solve the same forecasting problem.

The point was to compare different modeling philosophies, not just different hyperparameters.

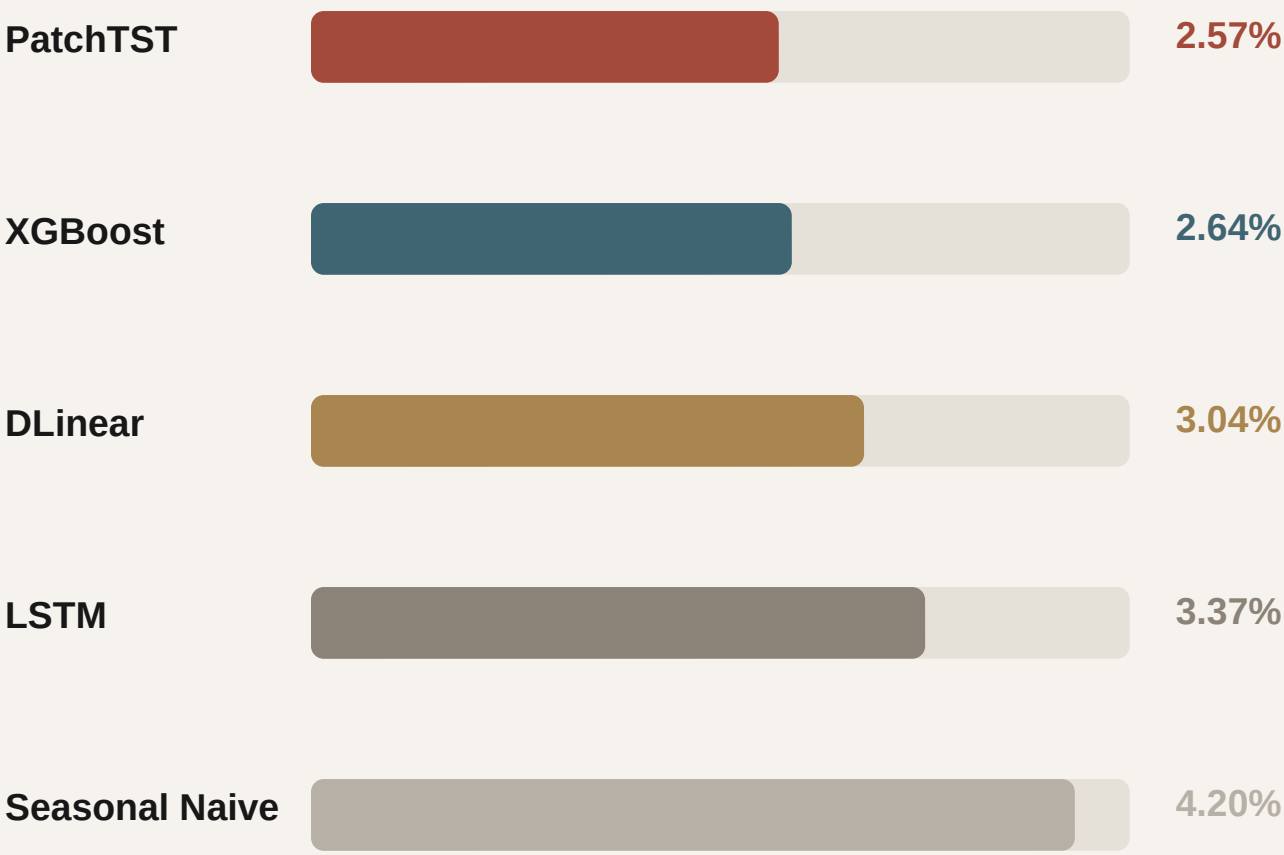
01	**Seasonal Naive**	**Rule-based baseline**	Same hour from the previous week
02	**XGBoost**	**Gradient-boosted trees**	Lag + calendar features
03	**DLinear**	**Lightweight neural model**	Linear trend/remainder mapping
04	**LSTM**	**Recurrent neural network**	Sequential hidden-state memory
05	**PatchTST**	**Transformer-based model**	Patches + self-attention

The comparison tested whether added architectural complexity actually improved the forecast.

FINAL TEST RESULTS

PatchTST won. XGBoost was almost tied.

WAPE on the 2020 test period · lower is better



Only ~0.07 percentage points separated PatchTST and XGBoost.

WHAT THE RANKING ACTUALLY SAYS

More complex did not mean more accurate.

Three comparisons mattered more than the final ranking itself.

XGBoost 2.64% vs **LSTM** 3.37%

Traditional ML beat the recurrent neural network.

DLinear 3.04% vs **LSTM** 3.37%

The simpler neural model beat the more complex one.

PatchTST 2.57% vs **XGBoost** 2.64%

The Transformer won, but only narrowly.

THE ENGINEERING TRADEOFF

The best score and the best choice are not always the same.

If accuracy is the only objective, PatchTST wins. If the model also has to be easy to iterate on and maintain, the decision is less obvious.

PatchTST

Best accuracy

- Lowest MAE, RMSE and WAPE
- Learns representations directly from the sequence
- More architecture, tuning and training overhead

XGBoost

Best practical tradeoff

- Nearly the same test accuracy
- Easier feature-driven workflow
- Faster iteration and simpler maintenance

My read: the extra Transformer complexity earned a small accuracy gain. That matters when every bit of error counts, but it would not be worth the overhead in every system.

TAKEAWAY

Complexity should earn its place.

PatchTST finished first, but the project changed how I think about model selection more than it changed which model I would always choose.

XGBoost showed how far a strong traditional ML model can go.

DLinear beating LSTM showed that “deeper” is not the same as “better.”

PatchTST showed that the most complex model can win, but the margin still has to justify the cost.

Electricity Demand Forecasting · Accuracy vs. Model Complexity